HTML TABLES AND LAYOUT

Organize Data and Create Structured Web Pages

1. INTRODUCTION

- HTML (HyperText Markup Language) is used to create web pages.
- Tables help in displaying data in rows and columns.
- Layout techniques help in arranging different parts of a web page in a clean and organized way.

2. PROBLEM / REQUIREMENT

We want to create a simple college information web page that can:

- Show student details in a table.
- Have a clear layout with header, navigation menu, content area and footer.
- Be simple, clean and easy to navigate.

3. WHY USE TABLES AND LAYOUT ?

- Show data in a clear and organized way.
- Divide the web page into different sections.
- Makes the page user-friendly and easy to navigate.
- Useful for websites like student portals, college websites, dashboards, etc.

4. BASIC HTML TABLE

Displays simple data in rows and columns.

HTML CODE

```
<table border="1">
  <tr>
    <th>Roll No</th>
    <th>Name</th>
    <th>Marks</th>
  </tr>
  <tr>
    <td>1</td>
    <td>Krishna</td>
    <td>85</td>
  </tr>
  <tr>
    <td>2</td>
    <td>Femi</td>
    <td>78</td>
  </tr>
</table>
```

OUTPUT

Roll No	Name	Marks
1	Krishna	85
2	Femi	78

PARTS OF A TABLE

`<table>`	Creates a table
`<tr>`	Creates a row
`<th>`	Table header (bold)
`<td>`	Table data (cell)
`border`	Adds border to table

5. TABLE WITH ROWSPAN AND COLSPAN

Used to merge rows or columns for better layout.

HTML CODE

```
<table border="1">
  <tr>
    <th colspan="3">Student Details</th>
  </tr>
  <tr>
    <th>Roll No</th>
    <th>Name</th>
    <th>Marks</th>
  </tr>
  <tr>
    <td>1</td>
    <td rowspan="2">Krishna</td>
    <td>85</td>
  </tr>
  <tr>
    <td>2</td>
    <td>78</td>
  </tr>
</table>
```

OUTPUT

Student Details		
Roll No	Name	Marks
1	Krishna	85
2		78

COLSPAN AND ROWSPAN

- **colspan** – merges columns (horizontally)
- **rowspan** – merges rows (vertically)

6. LAYOUT USING HTML

Divides a web page into different sections.

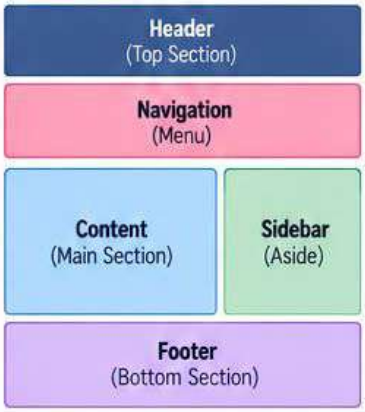
HTML CODE (Simple)

```
<div class="header">
  <h2>My College</h2>
</div>
<div class="nav">
  <a href="#">Home</a> |
  <a href="#">Students</a> |
  <a href="#">About</a>
</div>
<div class="content">
  <h3>Welcome</h3>
  <p>This is the main content.</p>
</div>
<div class="footer">
  <p>© 2026 My College</p>
</div>
```

OUTPUT (Web Page Layout)



LAYOUT STRUCTURE



7. DESIGN CONSIDERATIONS

- Keep layout simple and clean.
- Use appropriate colors and spacing.
- Use semantic tags for better structure.
- Make it responsive (works on mobile too).
- Ensure easy navigation for users.

8. STEP-BY-STEP PROCESS

1. Plan the layout (header, nav, content, footer).
2. Write the HTML structure using and <table>.
3. Add table for student data.
4. Use rowspan and colspan if needed.
5. Check the output in a browser.
6. Improve design using CSS (optional).

9. REAL-WORLD APPLICATIONS

- College websites (student information)
- Result sheets
- Product lists (e-commerce)
- Dashboards and reports